

Assignment 6

Task 1:

a)

P	N	Q ₊	Q ₊₊₁
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	1

b)

P'	N'Q'	N'Q	NQ	NQ'
0	0	0	1	0
1	1	0	1	1

$$f(P, N, Q) = \underline{\underline{NQ + Q'P}}$$

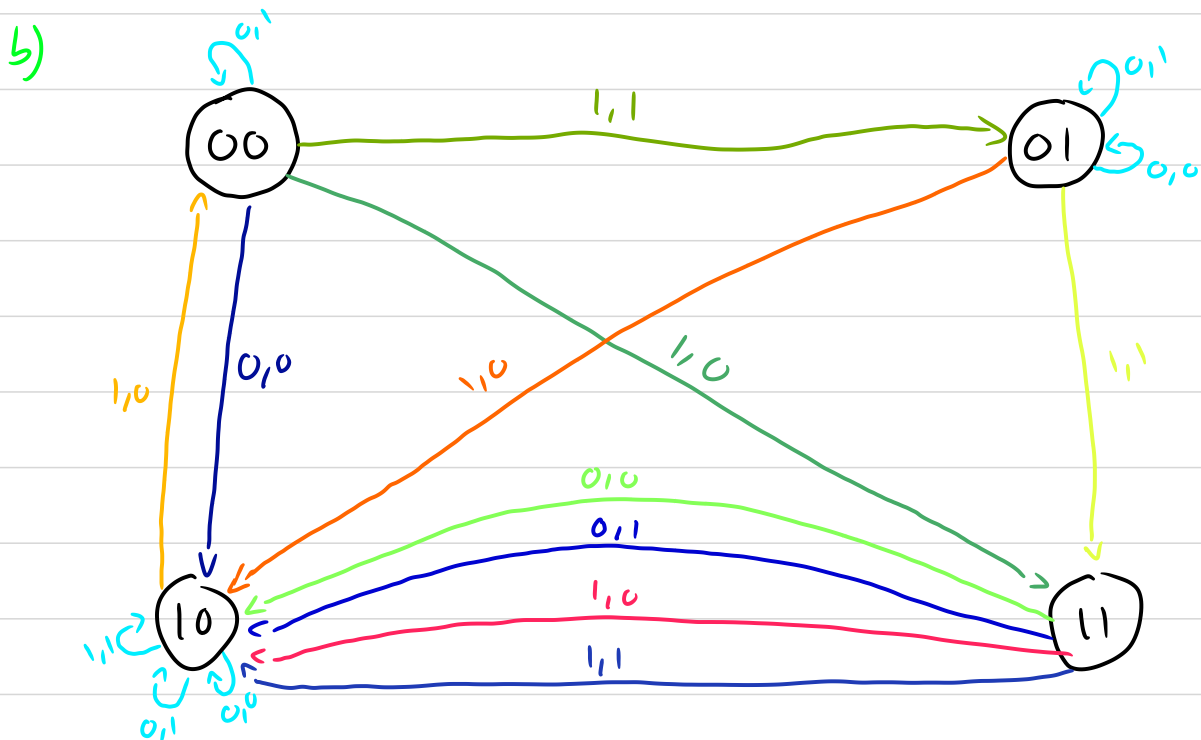
c)

Q(+)	Q(++1)	P	N
0	0	0	x
0	1	1	x
1	0	x	0
1	1	x	1

Task 2: *current*

2% current					transform				next		
a)	A	B	x	y	z	J _A	K _A	J _B	K _B	A ₊	B ₊
	0	0	0	0	0	1	0	0	0	1	0
	0	0	0	1	0	0	0	0	0	0	0
	0	0	1	0	0	1	1	1	1	1	1
	0	0	1	1	0	0	0	1	0	0	1
	0	1	0	0	1	0	0	0	0	0	1
	0	1	0	1	0	0	0	0	0	0	1
	0	1	1	0	0	1	0	1	1	1	0
	0	1	1	1	0	1	0	1	0	1	1
	1	0	0	0	1	1	0	0	1	1	0
	1	0	0	1	0	0	0	0	1	1	0
	1	0	1	0	0	1	1	0	1	0	0
	1	0	1	1	0	0	0	0	1	1	0
	1	1	0	0	1	0	0	0	1	1	0
	1	1	0	1	0	0	0	0	1	1	0
	1	1	1	0	0	1	0	0	1	1	0
	1	1	1	1	0	1	0	0	1	1	0

A/B	<i>transf.</i>		<i>next</i>
Q _t	J	K	Q _{t+1}
0	0	x	0
0	1	x	1
1	x	1	0
1	x	0	1



Task 3:

A	B	E	F	J _A	K _A	J _B	K _B	A(t+1)	B(t+1)
0	0	0	0	0	x	0	x	0	0
0	0	0	1	0	x	0	x	0	0
0	0	1	0	1	x	1	x	1	1
0	0	1	1	0	x	1	x	0	1
0	1	0	0	0	x	x	0	0	1
0	1	0	1	0	x	x	0	0	1
0	1	1	0	0	x	x	1	0	0
0	1	1	1	1	x	x	1	1	0
1	0	0	0	x	0	0	x	1	0
1	0	0	1	x	0	0	x	1	0
1	0	1	0	x	1	1	x	0	1
1	0	1	1	x	0	1	x	1	1
1	1	0	0	x	0	x	0	1	1
1	1	0	1	x	0	x	0	1	1
1	1	1	0	x	0	x	1	1	0
1	1	1	1	x	1	x	1	0	0

J	K	Q(t+1)
0	0	Q(t)
0	1	0
1	0	1
1	1	Q'(t)

Q(t)	Q(t+1)	J	K
0	0	0	x
0	1	1	x
1	0	x	1
1	1	x	0

J _A	A'B'	A'B	AB	AB'
E'F'	0	0	0	1
E'F	0	0	1	0
EF	x	x	x	x
EF'	x	x	x	x

$$F = AB'F' + ABF$$

K _A	A'B'	A'B	AB	AB'
E'F'	x	x	x	x
E'F	x	x	x	x
EF	0	0	1	0
EF'	0	0	0	1

$$F = AB'F' + ABF$$

K _A	A'B'	A'B	AB	AB'
E'F'	0	0	1	1
E'F	x	x	x	x
EF	x	x	x	x
EF'	0	0	1	1

$$F = AE' + AE$$

$$= A$$

K _B	A'B'	A'B	AB	AB'
E'F'	x	x	x	x
E'F	0	0	1	1
EF	0	0	x	1
EF'	x	x	x	x

$$F = AE' + AE$$

$$= A$$

$$F = \underline{AB'F' + ABF} \quad J_1 K_4$$

$$F = \underline{AB'F' + ABF}$$

$$F = AE' + AE$$

$$\underline{= A} \quad J_B K_B$$

$$F = \underline{AE' + AE}$$

